DA5400W: FML

Tutorial (Python) on Regression

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The following PDF includes a short recap along with some questions. Solutions/hints to specific problems have been added.

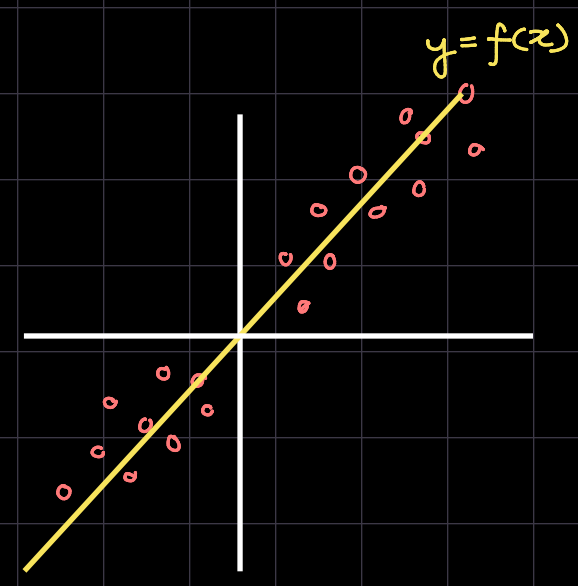
Linear regression :

$$y_i(x, w) = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$

In matrix form,

$$\underset{n\times 1}{y} = \underset{n\times d}{X}\,\underset{d\times 1}{w} + \varepsilon$$

random noise.
$E[\varepsilon]=0$, Gaussian



OLS: Ordinary Least Squares

$$SSE = \|y - Xw\|^2$$

$$SSE = (y - Xw)^T (y - Xw)$$

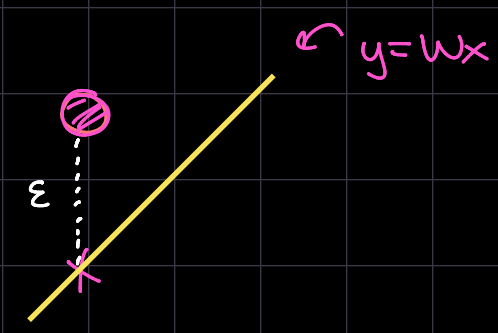
$$\hat{w} = \arg\min_{w} SSE$$

$$\frac{\partial}{\partial w} SSE = -2X^T(y - Xw)$$

set $\frac{\partial}{\partial w} SSE = 0 \Rightarrow X^T y - X^T X w = 0$

$$\Rightarrow X^T X w = X^T y$$

$$\Rightarrow \underline{w = (X^T X)^{-1} X^T y}$$

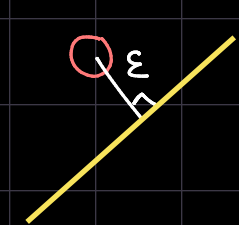


TLS: Total Least Squares

OLS assumes that there is noise only in y. TLS assumes there's noise in both X and y. so you find corrections in both X and y.

$$y + \Delta y = (X + \Delta x) w$$

It minimizes the orthogonal distances to the line.



Steps :

$A = [X \;\; y]$ ← Augmented Matrix

$A = U \Sigma V^T$ ← SVD

$v = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$ ← right singular vector corresp. to the smallest singular value.

$w = -\frac{v_x}{v_y}$

Essentially

OLS:

```
w = np.linalg.inv(X.T @ X) @ X.T @ y
```

TLS:

```
A = np.concatenate((X, y[..., None]), axis=-1)
U, s, Vt = np.linalg.svd(A, full_matrices=False)
w = -1 / Vt[..., -1, -1] * Vt[..., -1, :-1]
```

$A = [X \quad y]$

$SVD(A)$

Logarithmic Regression

$$y = a + b\ln x$$

$$x' = \ln(x)$$

$$y = a + bx'$$

Exponential Regression

$$y = ae^{bx}$$

$$\ln y = \ln a + bx$$

let $y' = \ln(y)$ $\quad a' = \ln(a)$

$$y' = a' + bx$$

Which of the following equations can be reformulated as linear regression models, treating (x, y) as variables and (a, b) as parameters? For each valid case, show the algebraic transformation that converts the equation into linear regression form.

(a) $y = ae^{bx}$

(b) $y = \frac{ax}{b+x}$

(c) $y = \frac{a}{x+b}$

(d) $y = a(1 - e^{-bx})$

Data on the monthly sales (y) in rupees and advertising spend (x) in rupees for 25 retail stores were collected. The mean and covariance matrix (about the mean) computed from the data is given:

$$z = \begin{bmatrix} y \\ x \end{bmatrix}, \quad \bar{z} = \begin{bmatrix} 200 \\ 40 \end{bmatrix}, \quad S_z = \begin{bmatrix} 1800 & 900 \\ 900 & 500 \end{bmatrix}$$

Assume a linear relation $y = ax + b$ exists between monthly sales (y) and advertising spend (x). Compute the model parameters using OLS approach.

Consider a linear regression model trained to predict y using predictors x_1 and x_2. Suppose the model is first trained with x_1 measured in kilometers (km). Now the unit of x_1 is changed to meters (m), so that

$$x_1' = 1000\, x_1.$$

The model is retrained using the transformed feature x_1'.

(a) For Ordinary Least Squares (OLS), explain how the coefficient of x_1 changes after rescaling. Will the model predictions change? Briefly justify.

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$y = ax + b$

$(y - \hat{y})^2$

$\varepsilon_i = (y_i - ax_i - b)^2$

$E = \sum_{i=1}^{n} (y_i - ax_i - b)^2$

$\frac{\partial E}{\partial b} = 0$

$\Rightarrow \bar{y} - a\bar{x} - b = 0$

$b = \bar{y} - a\bar{x}$

$\frac{\partial E}{\partial a} = 2\sum(\quad)(-x_i) = 0$

$\sum x_i(y_i - ax_i - b) = 0$

$\sum x_i(y_i - ax_i - \bar{y} + a\bar{x}) = 0$

$\Downarrow$

$\sum(x_i - \bar{x})(y_i - \bar{y}) = a\sum(x_i - \bar{x})^2$

$$\boxed{a = \frac{\mathrm{cov}(x, y)}{\mathrm{var}(x)}}$$

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$$\boxed{w_0 + \frac{1}{1000} w_1 \, 1000 x_1 + w_2 x_2}$$

$$w_1 x_1 + w_2 x_2$$

$$\hat{y} = w_0 + \left(\frac{w_1}{1000}\right)(1000 x_1) + w_2 x_2$$

(a) For Ordinary Least Squares (OLS), explain how the coefficient of x_1 changes after rescaling. Will the model predictions change? Briefly justify.

$$X' = XD$$

$$w = (X^T X)^{-1} X^T y$$

$$\hat{y} = Xw$$

$$w' = [(XD)^T (XD)]^{-1} (XD)^T y$$

$$= (D^T X^T X D)^{-1} D^T X^T y$$

$$= D^{-1} (X^T X)^{-1} \underbrace{(D^T)^{-1} D^T}_{I} X^T y$$

$$= D^{-1} (X^T X)^{-1} X^T y$$

$$w' = D^{-1} w$$

$$\hat{y} = Xw$$

$$\hat{y}' = X' w'$$

$$= XD D^{-1} w$$

$$= \boxed{Xw}$$